



100% Autonomous Cloud Atmospheric Intelligence & Mitigation

VayuVitals Engine V2 eliminates client-daemon dependencies by decoupling continuous atmospheric telemetry, Machine Learning inversion forecasting, and emergency dispatch into an authentic, serverless AWS Cloud pipeline running 24/7.

99

SPATIAL GRIDS

5km x 5km Delhi tessellation

332.4K

TRAINING SAMPLES

Delhi-NCR historical readings

30 Min

AUTONOMOUS CYCLE

Amazon EventBridge triggered

100%

ZERO-FAKING RIGOR

Real SageMaker + SES dispatches

THE PROBLEM: THE DELHI INVERSION TRAP

During post-monsoon and winter months, the boundary-layer in Delhi-NCR collapses to < 300m, trapping PM2.5, NO₂, and biomass smoke at human breathing levels.

Existing government apps report retrospective hourly averages **after** exposure occurs. School headmasters receive no forward warning before 7:00 AM school bus arrival, leading to hazardous outdoor morning assemblies and severe acute respiratory distress.

THE V2 BREAKTHROUGH: TRUE AUTONOMY

Engine V1 was bound to local server timers on client machines. If the operator closed their laptop, monitoring ceased.

Engine V2 migrates the entire pipeline into AWS Serverless. An independent Amazon EventBridge schedule invokes an AWS Lambda controller every 30 minutes, orchestrating real XGBoost inference on Amazon SageMaker and executing multi-channel SES dispatches completely off-grid.

V2 CLOUD ARCHITECTURE DATA PIPELINE



100% TRANSPARENCY & ZERO-FAKING DIRECTIVE

AGENTS.MD STANDARD ENFORCED

VayuVitals prohibits synthetic latency approximations, mock mathematical fallbacks (sine/cosine diurnal mocks), or fake cloud telemetry. Every metric, endpoint response, and email message ID presented is 100% verified against live Amazon Web Services APIs.



The 5km × 5km Spatial Grid Engine & The Three Pillars

Delhi's heterogeneous microclimates render regional city-wide AQI averages useless. VayuVitals discretizes Delhi-NCR into uniform geographic micro-cells to deliver localized actionable protection.

MICRO-SPATIAL GRID GEOMETRY

Latitude Bounds	28.40°N – 28.85°N (9 Rows)
Longitude Bounds	76.90°E – 77.40°E (11 Columns)
Step Discretization	0.05° Lat × 0.05° Lon Uniform
Physical Cell Size	5.54 km (N-S) × 4.88 km (E-W)
Total Discrete Blocks	99 Grids (GRID_R00_C00 – R08_C10)
Populated Institutional Blocks	60 Cells (Schools, Hospitals, Univs)

EMPIRICAL MICRO-TELEMETRY INGESTION

Rather than relying on isolated physical sensors with large spatial blind spots, VayuVitals aggregates continuous multi-parameter telemetry for each 5km block centroid:

- **Ground Parameters:** PM2.5, PM10, NO₂, CO, and Planetary Boundary Layer height.
- **Meteorology:** Temperature, Relative Humidity, Boundary Wind Vector.
- **Multi-Coordinate Ingestion:** Batch-queries 30+ block centroids in single network frames via Open-Meteo empirical API.

THE THREE PILLARS OF AUTONOMOUS INTERVENTION

PILLAR 1: PREDICTIVE MORNING FACILITY ADVISORIES

THRESHOLD: 75 $\mu\text{g}/\text{m}^3$ PM2.5

Target: School Principals, Headmasters, Clinic Directors, Hospital Administrators.

Function: SageMaker evaluates 48-hour forward hourly curves for each facility's exact GPS coordinates. If tomorrow's 7:00 AM – 9:00 AM arrival peak exceeds 75 $\mu\text{g}/\text{m}^3$, an operational advisory is issued specifying indoor assembly mandates and HVAC filtration directives.

Fatigue Suppression: If the forecasted peak is $\leq 75 \mu\text{g}/\text{m}^3$, the email is strictly suppressed to avoid alert fatigue. Single daily IST debounce prevents duplicate morning dispatches.

PILLAR 2: RAPID 5KM × 5KM BLOCK EMERGENCY SURGE ALERTS

THRESHOLD: 105 $\mu\text{g}/\text{m}^3$ PM2.5

Target: All institutional facilities situated inside the affected 5.5km × 4.88km spatial polygon.

Function: Scans all 99 spatial blocks every 30 minutes. If real-time particulate density in any block suddenly surges $\geq 105 \mu\text{g}/\text{m}^3$ (stubble smoke incursions or micro-inversion traps), an immediate emergency flash alert is dispatched.

Protocol: Mandatory outdoor recess suspension, exterior door/window sealing, and indoor HEPA purifier overdrive. Protected by a strict 3-hour debounce cooldown per grid block.

PILLAR 3: 14-DAY CHRONIC STATUTORY SECTION 10 LEGAL DOSSIERS

AIR (PREVENTION & CONTROL) ACT 1981

Target: Citizen Coalitions, School Legal Boards, DPCC, Municipal Corporation of Delhi (MCD).

Function: Continuously evaluates a rolling 14-day history (336 hourly entries) for every block. If a grid experiences 14 continuous calendar days of persistent severe statutory violations, VayuVitals compiles an evidentiary legal dossier.

Remedy: Empowers institutions under Section 10 of the Air Act 1981 to legally compel local authorities to deploy mechanical mist-cannoning, enforce road dust suppression, and divert heavy diesel transit corridors.



Machine Learning Validation & Production Telemetry

Prediction accuracy is the bedrock of institutional trust. The VayuVitals ML pipeline operates on high-dimensional gradient-boosted decision trees trained exclusively on real-world Indian atmospheric datasets.

SAGEMAKER MODEL PERFORMANCE (XGBOOST V4)

0.8156

R² EXPLAINED VARIANCE
Strong fit on non-linear peaks

27.92

TEST MAE (MG/M³)
Strict error tolerance

42.92

TEST RMSE (MG/M³)
Robust against outliers

76.8%

ACCURACY WITHIN ±40
54.3% within ±20 µg/m³

14 ENGINEERED INPUT FEATURES:

pm25_now horizon_hours (1-48) temperature humidity wind_speed hour_sin hour_cos doy_sin doy_cos
is_winter is_stubble_burning is_school_rush latitude longitude

VERIFIED LIVE AWS PRODUCTION AUDIT (ZERO-FAKING)

AWS SERVICE / RESOURCE	RESOURCE IDENTIFIER / ARN	OPERATIONAL STATUS & VERIFIED OUTPUT
Amazon EventBridge	wmd-autonomous-monitor-cron	ENABLED Fires rate(30 minutes) cloud cron
AWS Lambda	wmd-backend (ap-south-1)	IN-SERVICE Node 22, 512MB RAM, Timeout 180s
Amazon SageMaker	wmd-delhi-48h-forecast-endpoint	IN-SERVICE Serverless XGBoost inference
Amazon DynamoDB	AirQualityReadings	ACTIVE Persistent 14-day rolling compliance state
Amazon SES Dispatches	us-east-1 Verified Hub	MsgId: 010001a11a187de0... (Modern School Barakhamba) MsgId: 010001a11a1885bd... (Modern School Vasant Vihar) MsgId: 010001a11a188d68... (DPS Mathura Road) MsgId: 010001a11a189538... (DPS Rohini)

THE HORIZON: ENGINE V3 ROADMAP

1. Regional IGP Expansion

Expand the 5km grid tessellation across the entire Indo-Gangetic Plain (250+ blocks spanning Punjab, Haryana, UP, and Bihar).

2. Satellite AOD Fusion

Integrate INSAT-3DR and Sentinel-5P TROPOMI satellite Aerosol Optical Depth data directly into SageMaker feature matrices.

3. Automated Grievance Dispatch

Automate Pillar 3 dossier filings via API direct-connect with DPCC, MCD, and CPCB online grievance portals.